Experiment-17

Aim

To perform edge detection on an image using the Sobel operator along the X-axis in Python with OpenCV.

Procedure

1. Import the required libraries (cv2 and numpy).
2. Read the input image.
3. Convert the image to grayscale.
4. Apply the Sobel operator along the X-axis.
5. Convert the result to an 8-bit image.
6. Display the original image and the Sobel X output.
7. Save the output image if required.

Code:

```
import cv2

image = cv2.imread("sample.jpg")

if image is None:

   print("Error: Image not found!")

   exit()

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)



sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

sobel_x = cv2.convertScaleAbs(sobel_x)
```

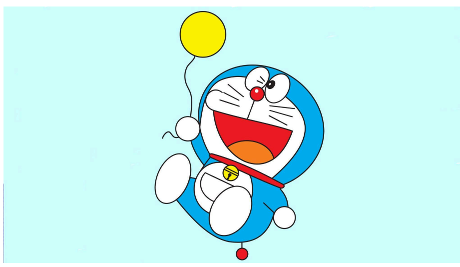
```
cv2.imshow("Original Image", image)

cv2.imshow("Sobel X Edge Detection", sobel_x)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Sample Input:



Output:

Results:

The Sobel edge detection algorithm along the **X-axis** was successfully applied to the input image. The resulting image clearly highlights the vertical edges by detecting intensity changes in the horizontal direction.